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Artificial bee colony optimization based energy-efficient wireless network interface selection for industrial mobile devices

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ABSTRACT

Management of energy is a crucial challenge for mobile devices, having the network activity frequently taking up a considerable part of the energy in the overall system. The important goal of this work is to offer energy efficiency for mobile handhelds. The industrial mobile devices in the recent times are fitted with several wireless network interfaces, these devices function with less battery power. In mobile handhelds, each user is able to run multiple applications with the purpose of have particular Quality of Service (QoS) requirements. Hence, it would be beneficial for a multi-interface terminal to simultaneously use two or more interfaces to gain in performance. So satisfying the QoS becomes very difficult task. An energy-effective Adaptive Wireless Network Interface Selection (AWNIS) with Artificial Bee Colony (ABC) is proposed in this paper. AWNIS-ABC algorithm chooses the best of the wireless network interface in regard to the energy consumed by taking the QoS of the link into consideration as well as following a hi-speed network interface-selection interval based on the scenario in the network. Then Hidden Semi Markov Model (HSMM) is introduced for searching the detouring paths if the energy state of a given relay node is very less to utilize the required interface, which ensures timely delivery. Also, it provides the diversification of the route paths in the network with the purpose of extending the lifetime of the network. An alternate path is proposed for reliable data delivery beforehand the route path encounters a breakage owing to energy exhaustion is found. A number of wireless network interfaces in mobile devices are utilized in improving the energy efficiency, satisfaction of QoS occurring during data transfer. The simulation of the network is implemented via the Network Simulator 2(NS2).

1. Introduction

With improved and sophisticated wireless technologies available, mobile devices attained considerable enhancement. Statistics [1] indicate that by 2014, in America, mobile devices would dominate the desktop devices and an average time of 2.7 h for each day is spent by people on their mobile devices. Users spent 89% of the time, on an average of 2.7 h for each day, on media via mobile APPs, whereas the respite of 11% media time via mobile web applications. Also, nearly 61% of mobile users play games that are energy-sensitive applications and are ruling over the overall application utilization. There are numerous energy sensitive applications exists, for instance, speech and object identification, face recognition, computer vision and graphics, natural language processing, virus scan, and mobile augmented reality.

In industrial environments, Mobile devices like Personal Digital Assistants (PDAs) as well as smart phones are utilized for controlling and monitoring machinery. For exchanging information amid workers, Mobile devices are as well utilized. These kind of mobile devices possess a number of wireless network interfaces, like 3G, Bluetooth, and Wi-Fi. Every wireless network interface contains considerably diverse features in regard to its energy utilization, data transfer rate, service location, and other aspects. Every wireless network interface contains diverse features in regard to its energy utilization, data transfer rate, service area and other aspects [2]. 3G networks offer extensive service areas as well as use up higher receive/transmission energy in comparison with Wi-Fi networks [3]. Usually, a Wi-Fi access point could enclose a radius of nearly 100~200 m. The 3G energy consumption is nearly twofold as close to the energy that is consumed by Wi-Fi for data transmission [2,3].

In mobile handhelds, efficient power management strategies are predominant, with the availability of ample mobile applications. Research and industrial groups have explored the aspects that affect the energy utilization in mobile devices as well as the methods for measuring the contribution made by those facets. Consider the mobile applications needing a greater processing power. For this case, computation offloading is found to be a common solution given for mobile devices by means of using their connectivity to the external

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computational resources through wireless communication technologies, for instance Bluetooth, 3G, 4G, or Wi-Fi. Fourth Generation (4G) mobile system will be moving forward in the next few years. From the users' perspective, the features of the 4G network systems will have a greater usability, greater energy efficiency, greater battery usage for providing support for multimedia services at lesser transmission expense, customization, and integrated services. As a result, the external computational resource runs the application and then transmits the outcomes of executing the application back on the mobile device. The main objective of computation offloading is to reduce the energy utilization and/or enhance application performance achieved in the mobile devices in comparison with the application that has to be run on the mobile device locally deprived of considering the capacities of other resource-abundant computing platforms. Even though, computation offloading offers larger benefits in energy efficacy as well as application performance, still there are few challenges blocking the construction of a more effective offloading framework.

The most important decisions taken for computation offloading are the time to offload and the infrastructure to be use so as to help the mobile users in offloading as explained in [4]. There are four critical challenges to be considered in offloading algorithms. The first challenge is that the algorithm has to be designed with success in order to conserve energy and/or enhance the performance of mobile systems. In addition to the offloading algorithms, the expense of data transfer sees an increase when making use of 3G or 4G wireless network technology. Subsequently, security is also regarded to be one among the significant challenges in offloading, as data will get transferred between mobile systems and other computational resources. One more challenge is the computing servers' availability. Otherwise said, there might a condition of unavailability of inaccessibility of computing servers for offloading destination in a public location. Amongst these, energy efficiency is regarded to be the critically significant challenge and therefore it is highlighted in this research.

Numerous researches were carried out to enhance as well as prolong the operational time of mobile handhelds by having a leverage over several network interfaces [5,6]. In order to develop resourceful wireless network interface selection approaches, various factors has to be taken into consideration, like the signal strength, kinds of the rate of data transfer, and the kinds of existing network interfaces.

On the one hand, each mobile terminal can simultaneously run different types of applications, which lead to the diverse flows with different Quality of Service (QoS) requirements in terms of throughput, delay, jitter, and packet loss. On the other hand, a multi-homed terminal is equipped with several interfaces; hence, it is possible to simultaneously use various available network interfaces and not simply to switch from a single network to another one. The earlier solution to this issue is majorly concentrated on solving network interface selection. But how to satisfy QoS requirements becomes major difficult issue. Hence, an effective network interface selection approach is necessary for enhancing the energy efficiency and solving QoS in network interfaces simultaneously.

In this work, suppose a mobile device used in the industrial setting is fitted with two wireless network interfaces, such as 3G, and Wi-Fi. Also, analysis on the delay involved in the data transfer of three diverse network interface-selection rules including the usage of Wi-Fi only, 3G only, and both Wi-Fi and 3G are carried out. An energy effective adaptive wireless network interface-selection approach that is termed Adaptive Wireless Network Interface Selection (AWNIS) with Artificial Bee Colony (ABC) is introduced depending on the results of analysis. A number of wireless network interfaces in mobile devices are utilized in improving the energy efficiency, and satisfaction of QoS occurring during data transfers. The objective is to select the association that best satisfies all QoS requirements of the network interface and efficient use of energy. The simulation of the network is also carried out with the help of Network Simulator 2(NS2). Depending on the results of simulation, it can be concluded that newly introduced AWNIS-ABC can enhance the energy efficiency as well as when ensuring a particular degree of data transfer delay, prolong the operational time of mobile devices. The organization of the paper work is described as follows: Section 2, the works relevant to the research issue is clearly discussed, Section 3 studies the proposed approach, Section 4, experimental results are discussed and finally, Section 5 deals with the conclusion and scope of the future work.

2. Related work

The energy aware resource management middleware structural design is Self-Tuning Power Manager (STPM) [7] and this is mainly concentrated on Input /Output (I/O) devices. It means wireless interfaces. This STPM is control among applications and systems and also it produces the system of energy sensitive caching. This application theory exposes the clues about the purposes after the I/O devices usage whereas the system discloses the contextual information on the work. It has been recognized and it is involved in a forthcoming problem of this journal. The information is final as it is represented, but the pagination of the overall state of the system to the applications. The power management methodology is used by the STPM with this information for the design concentrated from the applications. This is achieving the better relationship of performance and energy conservation with a particular application assignment.

Pering et al. [8] proposed the Cool Spots. This minimizes the energy consumed by the wireless mobile devices by determining when and whether to make use of Wi-Fi and Bluetooth in accordance with the difficulties in bandwidth. The system and rules are not required to make any alterations made to the mobile applications by themselves, and differences wanted for the obtainable organization are very less.

Tan et al. [9] described a novel client-centric protocol. The name of client-centric protocol is Power Saving Mode (PSM) generally requires the support of the structural design and it is able to decline the transmission output so it is directing to an increase in the transmission delay. According to the output this is not appropriate for Quality of Service (QoS) applications. And also the Transmission Control Protocol (TCP) traffic redesigning probability into occasional bursts by using the similar average throughput since it remains the same to the server transmission rate like the user can 10PSM exactly carry out the estimation of the arrival time of packets and assigning the wireless interface with a reduced power mode in accordance is designated.

Agarwal et al. [10] described an energy management structural design. The name of energy management structural design is Cell2Notify. It is engage the cellular radio equipped on a smartphone to weaken the wireless Local Area Network (LAN) interface through received Voice over Internet Protocol (VoIP) with respect to acquire call redirected over the Wi-Fi interface. This system gives a control on the enhanced quality and energy-efficiency of Wi-Fi and it is mainly used for the data transmissions while decreasing the scanning cost. The simulations results demonstrate that it is feasible to protect approximately 70%–80% of the energy in comparison with the single radio scenario. In this methodology, the researcher did not enable the possible energy protection and it can be accomplished with that system.

Blue-Fi [11] describes the energy saving by revealing the Bluetooth radios and context information to perform the paging channel form for IEEE 802.11b standard. The results of the system are deployed in iPAQ Personal Digital Assistants (PDA's) assisted with Bluetooth radios and Cisco AirNet wireless networking cards shows that it is feasible to preserve the energy among 23% and 48% in contrast with the current IEEE 802.11b standard operating modes with fewer effects terminated the performance. It permits the system to go switch on and off the wireless interface according to being available or not.

Catnap [12] reducing the consumption of energy by permitting the wireless interface to hide for the period of the time of data transmissions. This is attained by determining the traffic to incorporate insignificant voids in among packets into larger sleep intervals, even

throughout the time data transmissions. The system imposes was approved out beneath real-time network surroundings. On the other hand, the performance may get cooperated beneath the general mobile traffic owing to the reduced transfer size of mobile traffic. The number of lessons has been approved out to improve the energy effectiveness with numerous wireless network interfaces available in mobile devices.

Kim et al. [13] learned the energy-efficient Adaptive Wireless Network Interface-Selection (AWNIS) methodology. The AWNIS methodology is introduced on the basis of the arithmetical designs of energy consumed and patterns of data transfer delay. The AWNIS methodology is select the better wireless network interface with concern to energy consumed by compelling the quality link deliberation and using a dynamic network interface selection interval based on the development of network. In this case all the methodologies have no direct participation in the procedure of numerous wireless network interfaces, weak signal measurement and routing problems.

Niyato et al. [14] introduced two methods, specifically, population evolution and reinforcement-learning methods for network selection. Network selection is performed depending on the population evolution be able to reach the evolutionary equilibrium quicker, it needs a centralized controller towards collect, process, and broadcast data regarding the users in the related service area. Results of this method are empirically investigated. The accuracy of the simulation results attained from the game model is assessed with using simulations.

Goudarzi et al. [15] solved the issue of wireless network selection via vertical handover, depending on the received data, via embedding the decision issue in a Markov Decision Process (MDP) with Genetic Algorithm (GA). GA is introduced to discover a set of optimal decisions with the purpose of assures the best trade-off among QoS depending on their priority level. Then, Improved Genetic Algorithm (IGA) with Simulated Annealing (SA) is combined for search and optimization issues in heterogeneous wireless networks. Results demonstrate with the purpose of selecting the SA rules would reduce the cost function, and also demonstrate that the IGA-SA algorithm might reduce the no. of unnecessary handovers, and thus stop the 'Ping-Pong' effect.

Salih et al. [16] presented an overview of the network selection and prediction issues. In adding together, examine a complete classification of associated theoretic methods, and also study the integration among these methods, discovering the best result of network selection and prediction issues. The optimal solution is able to fulfil the needs of the next creation wireless networks. Liang and Yu [17] proposed work, the entropy and the fuzzy-analytic hierarchy process (FAHP) are used towards compute the objective weight and subjective weight of network attributes correspondingly, by means of FAHP particularly being used towards compute the user preference values of services for candidate networks. At last, Simple Additive Weighting (SAW), Multiplicative Exponent Weighting (MEW), and Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) are used related towards network attribute utility values and weights towards compute the score of every candidate network.

3. Proposed methodology

In this segment, the cost of data-transfer energy as well as the analysis of the energy taken up by a mobile device is defined in accordance with the network interface-selection interval. Moreover, it as well evaluates the data transfer delay of the before mentioned network interface-selection rules, which include: utilizing Wi-Fi only, 3G only, and both 3G and Wi-Fi. A dynamic Adaptive Wireless Network Interface-Selection approach (AWNIS) with Artificial Bee Colony (AWNIS-ABC) assignment technique was presented depending on the outcomes of the analysis. A number of wireless network interfaces in mobile devices are utilized in improving the energy efficiency, satisfaction of QoS occurring during data transfer. Hidden Semi Markov Model (HSMM) technique is utilized for minimizing the energy consumption at every sensor node and extending the network lifespan while transmitting the data packets from a source node to a destination, and meeting the packet threshold limits at the same time. The overall representation of the proposed work is shown in Fig. 1.

This work is motivated by the system model, which is explained as below:

3.1. System model

Consider an industrial mobile device, which is facilitated by means of several wireless network interfaces by considering its energy cost. For wireless data transfer, a modest data-transfer energy model is defined. The data-transfer energy cost along with the expense of forming a connection as well as then converting megabytes of data is modelled, like this:

$$E\left(i_k\right) = E_e\left(i_k\right) + n \times E_t\left(i_k\right) \tag{1}$$

As given in (1), utilizing the total of the network establishment cost, $E_e\left(i_k\right)$ as well as the energy cost taken for the transfer 'n' megabytes, $E_t\left(i_k\right)$, the data transfer cost of the i_k network interface. With the purpose of enhancing the energy effectiveness as well as minimize the volume of energy consumed, it is required to carry out the route selection process for data transfers over every instance of multiple wireless network selection. Route selection methods considers the packet delivery deadline R as an input, in addition to the time D as well as transmission energy E via every interface $i_k \in \{1, \ldots, M\}$ (in which M refer to the amount of available interfaces) . D and E stand for the vectors of size M, and they are initialized in incremental order of transmission energy (i.e., $D_1 > D_2 > \cdots > D_j > \cdots > D_M$ and $E_1 < E_2 < \cdots < E_j < \cdots < E_M$). Every sensor node contains the topology info for every interface type i (Topology) and could compute the shortest path amid the source and destination. A sensor node uses Dijkstra's algorithm with the intention of accomplishing this, in which every edge available amid nodes present in the topology gets weighed with the Expected Transmission Count (ETX) [18] to be the per-hop cost via the respective link. Taking this info into account, every node present alongside the path takes a decision on the local routing to choose the interface and the next subsequent hop for satisfying the packet delivery threshold when reducing the complete energy consumed in the system.

The objective function with Quality of Service (QoS) associated to interface i_k is calculated as follows:

$$ob = Q_t(i_k) + E\left(i_k\right) \tag{2}$$

$Q_t\left(i_k\right)$ &$E\left(i_k\right)$ represent the QoS, and energy consumption utilities of using network interface i_k determined by solution a_i. Four QoS criteria, including Jitter in [ms], Packet Delivery Ratio in [PDR(%)] Packet Loss in [PER(%)], and Latency in [ms] are considered to measure the interface utility, where their requirements are defined according to the type of the corresponding class.

Jitter: Jitter in turn is the delay between two consecutive packets in that frame.

Packet Delivery Ratio (PDR): PDR can be measured as the ratio of number of packets delivered in total to the total number of packets sent from source node to destination node in the network. It is desired that maximum number of data packets has to be reached to the destination. As the value of PDR increases the performance of the network also increases.

Packet Loss Ratio (PLR): While loss determines the maximum amount of packets loss the stream can tolerate to provide good quality.

Latency: Latency can be referred as the time delay between message transmission and message arrival. Latency is an important constraint for time-critical applications, and needs to be minimized.

Data-Transfer Energy Cost Modelling With 3G and Wi-Fi

In order to create the model of the data-transfer energy incurred with several wireless network interfaces, consider an industrial mobile device configured with a key wireless network interface and an

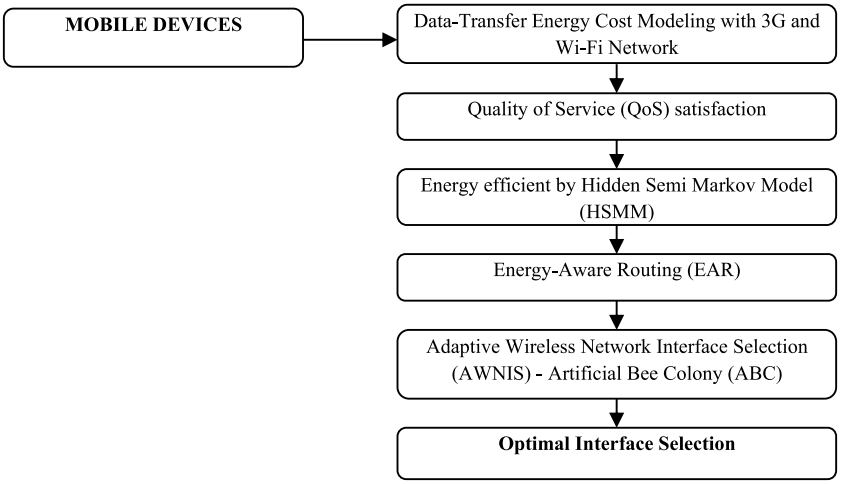


Fig. 1. Flow diagram of proposed system.

alternate wireless interface, called as a Smartphone. The mobile device can at all times link as well as transmission data by means of a low-energy as well as high-availability primary wireless network like 3G that renders a lesser data transfer rate and takes up more amount of energy in data-transfer in comparison with an alternate wireless network. As well, alternate wireless network interfaces like Wi-Fi are utilized with the intention of boosting the data transfer rate as well as energy effectiveness, for transferring the messages by having communication with a considerably near base station, like a Wi-Fi access point. If the device is connected to the alternate wireless network, the connection that is passing through primary wireless network gets disconnected. Primary wireless network interfaces takes up additional energy during data transfer compared to alternate wireless network interfaces owing to the long-range wireless communication. Therefore, compared to alternate wireless network interfaces, primary wireless network interfaces usually render lesser data transfer rates. The same kind of assumptions can be noticed in previous work [19] and [20]. Usually, the consumption of 3G is almost two times the volume of energy that is taken by Wi-Fi while transferring data [21]. In case, Wi-Fi is available in place of 3G in few locations, Wi-Fi has to be used for reducing the energy utilization of data transfers. In order to minimize the energy utilization as well as data transfer delay further, both 3G and Wi-Fi can be utilized alternatively by considering the availability of Wi-Fi. From the assumptions made above, the grade of Wi-Fi is presented based upon the Wi-Fi availability. In this, the state transition rate of Wi-Fi from the occupied state to the available state is defined. $\mu_{\mathrm{Wi-Fi}}$ refers to the state transition rate of Wi-Fi from the available state to the occupied one. $\lambda_{\mathrm{Wi-Fi}}$ stands for the state transition rate of Wi-Fi from occupied state to available state. Briefly stated, by means of a Markov process, the utilization patterns of 3G and Wi-Fi is modelled [22]. $\mu_{\mathrm{Wi-Fi}}$ as well as $\lambda_{\mathrm{Wi-Fi}}$ could vary based upon the network environment, on the extent of user movement, as well as on various other aspects.

As shown in (3), by computed the energy consumed during one network cycle, the energy utilization of a mobile device is analysed. $\mathrm{TC_{Device}}$ refers to the overall energy utilized by a mobile device for the period of one network cycle. The volume of energy depleted in the course of one network cycle consists of four important elements: the energy that is taken up by Wi-Fi, the energy taken up by 3G, the energy taken up by the choice of the network interface, in addition to the power taken up by the handover performed amid3G and Wi-Fi. The volume of power utilization for 3G and Wi-Fi are chiefly influenced by the volume of information that must be transmitted as well as the data transfer cost constant as the handover is carried out just once for each network cycle. Hence, the impact of the handover expenditure on the overall energy utilization is negligible, thus the vertical handover delay is minimized to close to zero based on the findings of previous research [23]. Hence, a means for reducing the energy consumption is expressed, as below:

$$TC_{Device} = TC_{transfer} + TC_{NIS} + TC_{HO} \tag{3}$$

$\mathrm{TC_{transfer}}$ stands for the total energy consumption pertaining to 3G and Wi-Fi for transferring the information in the course of one network cycle, $\mathrm{TC_{NIS}}$ refers to the overall energy consumed for the process of network interface selection per network cycle, $\mathrm{TC_{HO}}$ stands for the total energy utilized during the handover among Wi-Fi and 3G per network cycle.

In case the average requested data transfer quantity is considerably less, there is no necessity to utilize Wi-Fi as the network detection cost, that is to say the cost incurred in Wi-Fi scanning for finding an access point, is substantial. Hence, the volume of requested data that must be transmitted is taken into consideration. In the tth network cycle, $AR[t]$ represents the average requested data transfer amount (MB/s), e_{3G} and e_{Wi-Fi} stands for the data-transfer energy price of 3G and Wi-Fi per MB, correspondingly. $E[AR]$ Stands for the anticipated average amount of requested data transfer. The power utilization for each second of data transfer carried out by 3G and Wi-Fi are denoted by $e_{3G} \times E[AR]$ and, $e_{Wi-Fi} \times E[AR]$ correspondingly. The overall amount of energy taken up during the data transmission for one network cycle is computed in this manner:

$$\begin{aligned} TC_{transfer} &= e_{3G} \times E[AR] \times E\left[T_{3G}\right] + e_{Wi-Fi} \times E[AR] \times E\left[T_{Wi-F}\right] \\ &= e_{3G} \times E\left[T_{3G}\right] + e_{Wi-Fi} \times E[AR] \times E[AR] \end{aligned} \tag{4}$$

Periodic wireless network selection is necessary for selecting and connecting to the Wi-Fi network with more energy-efficiency. The overall energy utilization for the occasional wireless network selection is as below. Initially, the amount of network interface-selection events carried out with the aim of calculating the total amount of energy consumed during the network interface selection per network cycle is to be known. Every network interface selection is actually a Bernoulli trial. As described earlier, the outcome of every network interface selection either succeeds or fails. Hidden Semi Markov Model (HSMM) [24] is an effective probabilistic technique for measuring the probability values that Wi-Fi and 3G is existing subsequently t s by means that the transient state probability of a semi-Markov chain with a network interface-selection interval to reduce the consumption of energy as shown in Fig. 2. Hence, in addition to this, network interface-selection interval'd' of a provided state is specified for the HSMM. This period is actually a random variable as well as supposes an integer value existing in the set $\mathcal{TD} = \{1, 2, \ldots, \mathrm{D}\}$. The number of probability values that Wi-Fi and 3G is available i is decided by the duration of time

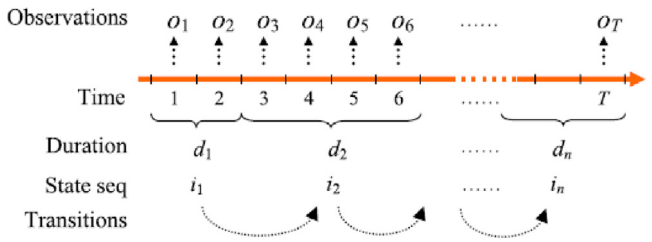


Fig. 2. HSMM depicted as a directed graphical model.

consumed during the state i, i.e., the time period. Suppose a discrete-time Markov chain with the group of (hidden) states $S = \{1, \ldots, M\}$, including available and unavailable (here M = 2). The state series is represented by $S_{1:T} \triangleq S_1, \ldots, S_T$, in which $S_t \in S$ refers to the state at particular time t. Generally HSMM is regarded to be two types of state sequences like hidden state and observed data state. In the hidden state, both the Wi-Fi and 3G network are either in available or unavailable state simultaneously. In the observed state, the total power consumption results of both the Wi-Fi and 3G networks is decided for network selection. Assume a hidden state sequence $S \triangleq \{Wi-Fi_1, Wi-Fi_2, \ldots 3G_{1T}\}$ and the observed data sequence results tends to become the results of energy consumption for every network. A model of $S_{1:T}$ is represented as s 1:T .

To simplify, in the next sections that follow,

- $S_{[t_1:t_2]} = Wi-Fi_{i[t_1:t_2]} = i$– state i (Available or unavailable) that the system remains in during the time period from t_1 to t_2 in $Wi-Fi_1$ to $Wi-Fi_2$ or $3G_1$ to $3G_2$. Otherwise said, it implies $S_{t_1} = i, S_{t_1-1} = i, \ldots$, and $S_{t_2} = i, \ldots$,. It is to be noted that the former state S_{t_1-1} as well as the subsequent state S_{t_2+1} may or may not be i.
- $S_{[t_1:t_2]} = Wi-Fi_{i[t_1:t_2]} = i$– state i (Available or unavailable) that begins at time t_1 and completes at t_2 with duration $d = t_2 - t_1 + 1$ in $Wi-Fi_1$ to $Wi-Fi_2$ or $3G_1$ to $3G_2$. This means that the former state S_{t_1-1} and the subsequent state S_{t_2+1} should not be i.
- $S_{[t_1:t_2]}$ = i – state i (Available or unavailable) that begins at time t_1 and continues till t_2, with $S_{[t_1]} = i, S_{t_1+1} = i, \ldots, S_{t_2} = i$ where $S_{[t_1]} = i$ implies that at t_1 the system moved from some other state to $Wi-Fi_1$ to $Wi-Fi_2$ or $3G_1$ to $3G_2$i, that is to say the former state S_{t_1-1} must not be i. The subsequent state S_{t_2+1} may or may not be i.
- $S_{[t_1:t_2]}$ = i – state i (Available or unavailable) that stays from t_1 to t_2 and ends at t_2 with $S_{t_1} = i, S_{t_1+1} = i, \ldots, S_{[t_2]} = i$ = i, in which $S_{[t_2]} = i$ indicates that at time t_2 the state would complete as well move to some other state, which is $Wi-Fi_1$ to $Wi-Fi_2$ or $3G_1$ to $3G_2$, that is to say the next state S_{t_2+1} should not be i. The former state S_{t_1-1} might or might not be i.

Depending upon these above mentioned definitions, $S_{[t]}$ = i implies state i commencing as well as ending at t with a time interval 1, $S_{[t]} = i$ implies state i beginning at t, $S_{[t]} = i$ indicates state i ending at t, and $S_t = i$ implies the state at t being state i. The energy consumption observation series is represented by $O_{1:T} \triangleq O_1, \ldots, O_T$, where $O_t \in \mathcal{V}$ is known as the observable at time t and $V = \{v_1, v_2, \ldots, v_K\}$ stands for the collection of observable energy consumption values. For energy consumption observation series $O_{1:T}$ the underlying state sequence is expressed by $S_{[1:d_1]} = i_1$, $S_{[d_1+1:d_1+d_2]} = i_2, \ldots, S_{[d_1+\cdots+d_{n-1}+1:d_1+\cdots+d_n]} = i_n$ and the state transitions are $(i_m, d_m) \rightarrow (i_m + 1, d_m + 1)$, for $m = 1, \ldots, n-1$, where $\sum_{m=1}^{n} d_m = T, i_1, \ldots, i_n \in S$, and $d_1, \ldots, d_n \in D$. It is to be noted that the first state i_1 is not required to start at time 1 corresponding to the first energy consumption observation o_1 and the final state is not required to end at time T corresponding to the last energy consumption observation o_T. The state transition probability from $(available, d') \rightarrow (unavailable, d)$ for $available \neq unavailable$ is defined by.

$$a_{(available,d')(unavailable,d)} \triangleq P\left[S_{[t+1:t+d]} = Wi-Fi_1 \text{ is unavailable after t seconds (j)} \middle| S_{[t-d'+1:t]} = Wi-Fi_1 \text{ is available after t seconds (i)}\right] \quad (5)$$

Depending on $\sum_{j \in S\{i\}} \sum_{d \in \mathcal{D}} a_{(available,d')(unavailable,d)} = 1$ with zero self-transition probabilities $a_{(available,d')(unavailable,d)} = 0$, in which available, unavailable $\in$ S and $d, d' \in D$. It is can be concluded from the definition that the former available state begun at $t - d' + 1$ and ended at t, with duration d'. After that it moves to state j (*unavailable*) with time interval d, based on the state transition probability $a_{(available,d')(unavailable,d)}$. Unavailable State j would begin at t + 1 and end at t + d. When in Unavailable state d energy consumption observations $o_{t+1:t+d}$ that are being emanated are defined. This emission probability is denoted by,

$$b_{Wi-Fi_1 \text{ is unavailable after t seconds,d}}(o_{t+1:t+d}) \triangleq P\left[o_{t+1:t+d} \middle| S_{[t+1:t+d]} = Wi-Fi_1 \text{ is unavailable after t seconds}\right] \quad (6)$$

That is supposed to be not dependent on time t. Consider the primary distribution of the state be

$$\pi_{WiFi_1 \text{ is unavailable after t seconds,d}} \triangleq P\left[S_{[t+1:t+d]} = Wi-Fi_1 \text{ is unavailable after t seconds}\right], \quad t \leq 0, d \in D \quad (7)$$

$$P\left(Wi-Fi_1 \text{ is unavailable after t seconds} \middle| \text{unavailable} \middle| \times \pi_{WiFi_1 \text{ is unavailable after t seconds,d}}, E\right) \quad (8)$$

$$= \frac{\mu_{Wi-Fi}}{\lambda_{Wi-Fi} + \mu_{Wi-Fi}} + \frac{\lambda_{Wi-Fi}}{\lambda_{Wi-Fi} + \mu_{Wi-Fi}} \times e^{-(\lambda_{Wi-Fi} + \mu_{Wi-Fi})t}$$

$$P\left(Wi-Fi_1 \text{ is available after t seconds} \middle| \text{unavailable} \middle| \times \pi_{WiFi_1 \text{ is unavailable after t seconds,d}}, E\right) \quad (9)$$

$$= \frac{\lambda_{Wi-Fi}}{\lambda_{Wi-Fi} + \mu_{Wi-Fi}} - \frac{\mu_{Wi-Fi}}{\lambda_{Wi-Fi} + \mu_{Wi-Fi}} \times e^{-(\lambda_{Wi-Fi} + \mu_{Wi-Fi})t}$$

After that the collection of the model parameters for the HSMM is stated as,

$$\lambda \triangleq \{a_{(available,d')(unavailable,d)}, a_{(unavailable,d)}\left(v_{k_1} : k_d\right), \pi_{WiFi_1 \text{ is available after t seconds,d}}\} \quad (10)$$

For a particular path $Wi-Fi_n$ adapting the observed energy consumption, the results are generated individually based on

$$p\left(TC_{energy} | Wi-Fi\right) = \prod_{n=1}^{N} p\left(TC_{energy_n} | Wi-Fi_n\right) \quad (11)$$

$p\left(TC_{energy} | Wi-Fi\right)$ is frequently referred to as the emission probabilities results for energy. One among the important issues occurring in the present HSMM techniques is about linear time efficiency. Once the probability values are got, a route selection approach is proposed, which takes the energy availability in nodes into consideration while taking route decisions. This Energy-Aware Routing (EAR) is proposed. But, from Algorithm 1, Energy-Aware Routing for Transfer and Receive the Packet (EARTR) has one more step to verify whether the chosen next hop contains sufficient energy to as well as transfer and receive the packet (lines 13–15). Otherwise, EARTR chooses an alternate next hop from its neighbourhood. Algorithm 1 renders the details for selecting alternate next hop. First, all the immediate neighbours of current Node that are approachable via interface type j is identified. After that the path as well as cost to destination is calculated by means of each one of the alternate next hop candidates. A random next hop candidate whose path cost ($costNxt_j$) is lesser compared to the path

cost of the current node ($cost_j$) (i.e., choosing a next-hop node nearer to destination instead of the present node) is selected. The last mentioned ensures a route that is loop-independent.

Algorithm 1: Energy-Aware Routing (EAR)

1. Input: $cost_j$, current Node, dst, D, TC_{energy}
2. Output: Alternative next hop
3. INeigh = FindImmediateNeighbors(currentNode, j)
4. for all nodes n in I Neigh do
5. for all interfaces j $\in \{1,..,M\}$ do
6. [pathNxt$_j$,costNxt$_j$] = Dijkstra's(next Node, dst, TC_{energy},)
7. Select arg $min_{j\in\{1,..M\}} TC_{energy}$,. s. t $N_j * D_j \leq R_{nextnode}$
8. $P_{txNextNode} = TC_{energy}$;
9. $P_{rxNextNode} = TC_{energy}$ {Next statement ensures a loop-free route}
10. if costNxt$_j$<cost$_j$ and energy map[next Node] <$P_{rxNextNode}$ + $P_{txNextNode}$ then
11. neighbors = neighbors: Append(n)
12. end if
13. end for
14. end for
15. {Choose a random alternate next hop from neighbors}
16. $altNextHop = SelectRandom(neighbors)$
17. return altNextHop

In case of Wi-Fi being in place, Wi-Fi could be utilized for transferring data. When δ_{NIS} is utilized in the form of the network interface-selection duration, then the probability that selection of a network interface becoming a success or failure could be computed in both scenarios using,

$$P_{NIS-Fail}\left(\delta_{NIS}\right) = \frac{\mu_{Wi-Fi}}{\lambda_{Wi-Fi}+\mu_{Wi-Fi}} + \frac{\lambda_{Wi-Fi}}{\lambda_{Wi-Fi}+\mu_{Wi-Fi}} \tag{12}$$

$$\begin{aligned} P_{NIS-Succ}\left(\delta_{NIS}\right) &= \frac{\mu_{Wi-Fi}}{\lambda_{Wi-Fi}+\mu_{Wi-Fi}} - \frac{\lambda_{Wi-Fi}}{\lambda_{Wi-Fi}+\mu_{Wi-Fi}} \\ &\quad \times e^{-(\lambda_{Wi-Fi}+\mu_{Wi-Fi})^{\delta_{NIS}}} \\ &= 1 - P_{NIS-Fail}\left(\delta_{NIS}\right) \end{aligned} \tag{13}$$

Therefore, the probable number of network interface-selection events carried out $E\left[N_{NIS}\right]$ can be computed by (14) using $P_{NIS-Succ}\left(\delta_{NIS}\right)$. $E\left[T_{3G}\right]$ is computed by (13) by employing $E[N_{NIS}(\delta_{NIS})]$ and δ_{NIS}:

$$E\left[N_{NIS}\left(\delta_{NIS}\right)\right] = \frac{1}{P_{NIS-Succ}\left(\delta_{NIS}\right)} \tag{14}$$

$$E\left[T_{3G}\right] = E[N_{NIS}(\delta_{NIS})] \times \delta_{NIS} \tag{15}$$

For $E[T_{Wi-Fi}]$, it can be computed easily with (14) provided the memory-less characteristic of an exponential distribution [25]:

$$E\left[T_{Wi-Fi}\right] = \frac{1}{\mu_{Wi-Fi}} \tag{16}$$

TC_{NIS} selections during one network cycle. TC_{NIS} varies based on the network interface-selection interval. If δ_{NIS} is utilized as the interval, the overall energy consumed during the network interface selection per network cycle, $TC_{NIS}(\delta_{NIS})$, can be computed using

$$TC_{NIS}\left(\delta_{NIS}\right) = e_{NIS} \times E\left[N_{NIS}\left(\delta_{NIS}\right)\right] \tag{17}$$

At last, the energy utilization of the mobile device could be calculated and estimated with 3G as well as Wi-Fi interfaces per network cycle employing all the equations studied earlier.

3.2. Delay modelling with 3G and Wi-Fi network interfaces

The data transfer delay is also regarded to be a significant challenge to be considered while assessing the efficacy of a data transfer. Here, in this section, the delay incurred in a mobile device fitted with 3G as well as Wi-Fi network interfaces is modelled. Delay may occur in two stages: Interference is one among the critical challenges in network interface-selection. The second step is concerned with the estimation of the anticipated delay depending on the volume of the data transfer requested. The delay is defined to be the time needed for transferring the message requested. Three data transfer policies are merely used, which include: the Wi-Fi-only, the 3G-only, and the 3G and Wi-Fi policies. The delay incurred for every policy is computed as below. The queuing delay for the requested data transfer is ignored in the delay analysis.

(1) Using 3G Only: While making use of the 3G-only policy, only 3G is used for transferring the data requested. The anticipated delay is calculated simply with the help of (18). In (18), $E[D_{3G}]$ represents the anticipated delay associated with the 3G-only policy as well as $E[R_{3G}]$ stands for the anticipated data transfer rate of 3G. BD, in units of MB, represents the volume of the data transfer requested or else the backlogged data. The anticipated delay incurred in 3G could just rise in linear fashion in accordance with BD and $E[R_{3G}]$. It can be easily predicted that the anticipated delay reduces as $E[R_{3G}]$ sees an increase:

$$E\left[D_{3G}\right] = \frac{BD}{E[R_{3G}]} \tag{18}$$

(2) Using Wi-Fi Only: While making use of the Wi-Fi-only policy, just Wi-Fi has to be used for transferring the data. This maximizes the energy efficacy of the data transfer, on the other hand there might be an increase in delay as the service region of Wi-Fi access points is less in few locations. The anticipated delay when making of Wi-Fi only, $E[D_{Wi-Fi}]$ is calculated by

$$\begin{aligned} E\left[D_{Wi-Fi}\right] = E\left[A_{Wi-Fi}\right] &\times \left(\left(NS_{Wi-Fi}\right) \times E\left[T_{3G}\right] + \frac{BD}{E\left[R_{Wi-Fi}\right]}\right) \\ &+ \left(1 - E\left[A_{Wi-Fi}\right]\right) \\ &\times \left(\left(NS_{Wi-Fi}\right) \times E\left[T_{3G}\right] + \frac{BD}{E\left[R_{Wi-Fi}\right]}\right) \end{aligned} \tag{19}$$

where

$$NS_{Wi-Fi} = \left\lceil \frac{BD}{\left(E\left[R_{Wi-Fi}\right] \times E\left[T_{Wi-Fi}\right]\right)} \right\rceil \tag{20}$$

In (20), NS_{Wi-Fi} represents the number of minimum network cycles necessary for transferring the desired data when making use of the Wi-Fi-only policy. In case a data transfer request comes up, then the network connection could be 3G or Wi-Fi. In case, the data transfer commences with a Wi-Fi connection, then the network cycle comprises of a series of Wi-Fi as well as 3G. Else, the network cycle comprises of a series of 3G as well as Wi-Fi. Hence, the accessibility of Wi-Fi has to be considered. The availability of Wi-Fi $E\left[A_{Wi-Fi}\right]$ is computed as

$$\left(\frac{E\left[T_{Wi-Fi}\right]}{E\left[T_{3G}\right]} + E\left[T_{Wi-Fi}\right]\right), \text{ or } \alpha \times A_{Wi-Fi}[t] + (1-\alpha) \times E\left[A_{Wi-Fi}\right] \tag{21}$$

where t refers to the network cycle index. $\left[A_{Wi-Fi}\right]$ is used for denoting the availability of Wi-Fi in the future.

(3) Utilizing Both 3G and Wi-Fi: Involves the usage of both 3G and Wi-Fi complementary for data transfer. When 3G is essentially used for transferring the data, by events involving regular wireless network interface-selection, Wi-Fi is as well utilized for transferring the data with the purpose of minimizing the volume of energy consumption as well as the delay that could be minimized considerably more while transmitting huge chunks of data. The delay expected while making use of both 3G and Wi-Fi, $E[D_{3G|Wi-Fi}]$, is computed by

$$\begin{aligned} E\left[D_{3G|Wi-Fi}\right] = E\left[A_{Wi-Fi}\right] &\left(\left(NS_{3G|Wi-Fi} - 1\right)\right. \\ &\left. \times \left(E\left[T_{3G}\right] + E\left[T_{Wi-Fi}\right]\right) + \beta_1\right) \end{aligned} \tag{22}$$

$$+\left(1-E\left[A_{Wi-Fi}\right]\right)\times\left(\left(NS_{3G|Wi-Fi}-1\right)\times\left(E\left[T_{3G}\right]+E\left[T_{Wi-Fi}\right]\right)+\beta_1\right)$$

where

$$NS_{3G|Wi-Fi}=\frac{BD}{\left(E\left[R_{3G}\right]\times E\left[T_{3G}\right]\right)+\left(E[R_{Wi-Fi}]\times E[T_{Wi-Fi}]\right)} \tag{23}$$

In (23), $NS_{3G|Wi-Fi}$ represents the number of network cycles necessary for transferring the desired data while utilizing 3G as well as Wi-Fi. A request for data transfer could come up in 3G or Wi-Fi similar to the Wi-Fi-only policy. Hence, the availability of Wi-Fi has to be considered while computing the delay:

$$\text{If } \left[\frac{RD}{E[R_{Wi-Fi}]\times E[T_{Wi-Fi}]}\right]\geq 1 \tag{24}$$

$$\text{Then } \beta_1=E\left[T_{Wi-Fi}\right]+\frac{\left(RD-\left(E[R_{Wi-Fi}]\times E[T_{Wi-Fi}]\right)\right)}{E[R_{3G}]} \tag{25}$$

$$\text{Else } \beta_1=\left[\frac{RD}{E\left[R_{Wi-Fi}\right]}\right] \tag{26}$$

$$\text{If } \left[\frac{RD}{E[R_{3G}]\times E[T_{3G}]}\right]\geq 1 \text{ Then } \beta_2=\frac{RD}{E[R_{3G}]} \tag{27}$$

where

$$RD=BD-\left(NS_{3G|Wi-Fi}-1\right)\times\left(E\left[R_{3G}\right]\times E\left[T_{3G}\right]+E[R_{Wi-Fi}]\times E[T_{Wi-Fi}]\right) \tag{28}$$

The amount of $E\left[R_{3G}\right]\times E\left[T_{3G}\right]+E[R_{Wi-Fi}]\times E[T_{Wi-Fi}]$ data is transmitted for each network cycle. For the period of the final network cycle of the data transfer, 3G or Wi-Fi or both 3G and Wi-Fi is utilized for transferring the remaining data. It is required to compute the delay incurred during the final network cycle individually. At this point, β_1 and β_2 represents the delays probable for the in all network cycle for the two cases. First, β_1 and β_2, which decides the volume of Residual Data (RD) that has to be transmitted for the period of the final network cycle has to be calculated. In case RD is higher than the amount of data, which is transmitted by the first network that is utilized (3G or Wi-Fi) for the period of the final network cycle, after that 3G as well as Wi-Fi is utilized for transferring the residual data for the period of the final network cycle. Else, only 3G or Wi-Fi are used for transferring the residual data.

Artificial Bee Colony (ABC)

Artificial Bee Colony (ABC) resembles the foraging behaviour of honey bees [26]. When the bee tries optimizing the energy value in Wi-Fi and 3G, they generally deposit a distinct dancing characteristic of the bees. In ABC, al the employee bees collect information about energy values in Wi-Fi and 3G. Employed bees find the local information of energy values in Wi-Fi and 3G and try to get the results of optimal energy values locations for Wi-Fi and 3G in the selection of multiple network interfaces. The bees that are waiting in the nest area for the discovery of most of the energy values for Wi-Fi and 3G in multiple network interface selection are considered to be onlooker bees. Onlooker bees perform the global analysis of energy values in Wi-Fi and 3G and thereafter revise the global energy values results during the position update stage. Scout bees decide on the randomly selected new energy values in Wi-Fi and 3G that is not searched by the employed bees, these three phases are preceded until a greatest number of the iterations termination condition is satisfied. The fitness value is calculated based on Eq. (28). An artificial onlooker bee selects according to the computation of the probability value p_a by employing expression provided below,

$$p_a=\frac{RD}{\sum_{w=1}^{SN}RD\left(E\left[R_{3G}\right]\times E\left[T_{3G}\right]+E[R_{Wi-Fi}]\times E[T_{Wi-Fi}]\right)} \tag{29}$$

where RD specifies the fitness value to each employee i in the location belong to the range between 0 to 1 which is computed from Eq. (2). and SN indicates the size of the population. The selected channels position is revised by making use of the below Eq. (30)

$$v_{ab}=Ew_{ab}+\theta_{ab}\left(E_{ab}-\chi_{cb}\right) \tag{30}$$

where c and b are randomly selected energy values $c\in\{1,2,\ldots,SN\}$ & $b\in\{1,2,\ldots,D\}$. $\phi_{ab}\in[-1.1]$, depending on this result, the parameter value of E_{ab} goes beyond its threshold value, and the result of the bees for Wi-Fi and 3G is satisfying. Else if it is not satisfying, it is replaced by the scouts bees, In ABC, if a bee present energy values position does not help in improving the result within a pre-defined number of iterations, and next the current energy values to be assumed as abandoned and it is updates as given below,

$$EW_a^b=E_{min}^b+rand\,(0,1)\left(E_{max}^b-E_{min}^b\right) \tag{31}$$

All the above mentioned steps are mainly dependent on the below parameters that restricts the operation SN, which is the maximum number of the cycles (MNC).

Algorithm 2: Energy estimation using ABC

Input: SN represent the size of bees, which is number of energy values, Drepresents the dimensions of search space for energy values

Output: Update energy values for Wi-Fi and 3G channels

Method

1. Initialize the energy values of bee with random positions and velocities v_{ab} in the search space$D$, and then choose the most worthy optimal energy values
2. Fix $cycle=1$
3. Repeat
4. For the complete energy values in bee do
5. Estimate the objective function from RD and ob
6. Build new energy values computation for every bee position v_{ab}in (31)
7. Execute the greedy selection procedure for the employed bees
8. Calculate the probability values $p_a$for select energy values in (30)
9. Build the current energy values position solutions v_a for the onlookers bees based on P_i and assess them
10. Realize the greedy selection technique for the onlookers
11. Decide the discarded tasks results in the scout, and if it is available, substitute it with a new randomly generated solution $EW_a^b$by (32)
12. Remember the best solution achieved to this point
13. $cycle=cycle+1$
14. Until $cycle=MCN$is met (generally sufficiently better energy values fitness)
15. Retrieve the best energy values and its fitness value
16. The updated energy values by means of their respective energy values to get the new Wi-Fi and 3G channels
17. Establish data transmission.

4. Experimentation results

In this section, a simulation is performed in NS2 simulator for the evaluation of AWNIS, AWNIS-PSO and AWNIS-ABC. The network field is segregated into Wi-Fi service regions and 3G service regions. 3G encloses the entire network area. The service regions of Wi-Fi are less. A user having a mobile device travels around inside the network area. The user movement differs between 1 m/s and 10 m/s. Table 1 illustrates the parameters employed in the simulation process. The results of simulation are obtained after each simulation being performed 100 times to receive the mean energy consumption and the data transfer delay.

The performance of AWNIS, AWNIS-PSO and AWNIS-ABC is evaluated in comparison with the below five network interface-selection policies in terms of the consumption of energy and data transfer delay. Moreover, two better policies, which make the optimal use of 3G and Wi-Fi and Wi-Fi-only policies are defined. Suppose that the exact state of Wi-Fi is known information for these two optimal policies. These two policies can vary the network interface to Wi-Fi at the precise time moment when Wi-Fi is available. Hence, occasional network interface selection is not necessary in these two policies and its expenditure

Table 1
Simulation parameters.

Parameters	Value
Network field size	1000 m × 1000 m
Workloads	10, 50, 100, 150, 200 kB/s, ECG, realistic traffic
Energy cost of 3G for data transferring	28.5 J/MB
Energy cost of Wi-Fi for data transferring	12.9 J/MB
Energy cost of a network selection interval	5.9 J
Initial network interface selection interval	30 s
Maximum network interface selection interval	300 s
Wi-Fi	802.11 g
3G	HSPA
The number of access points	10, 15, 20, 25
The coverage of an access point	100 m rad
User mobility	1 m/s–10 m/s

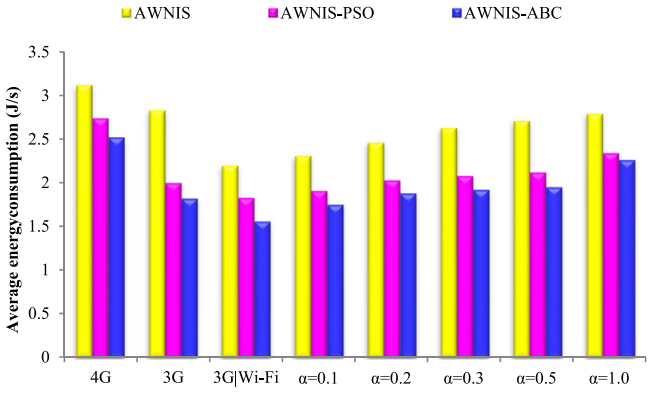


Fig. 3. Average energy consumption comparison (I) as per the coefficient value α, where the number of APs is 15.

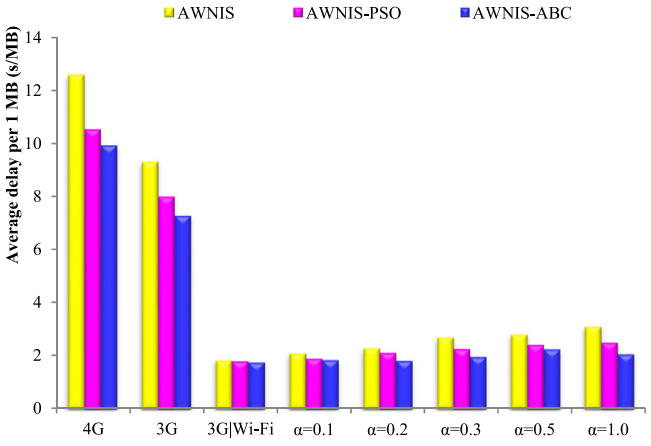


Fig. 4. Average data transfer delay comparison (I) as per the coefficient value, where the number of APs is 15.

does not get accounted for the energy consumption, and Wi-Fi can be used to the maximum degree possible but this supposition is not practical in the real time systems. This being said, the two optimal policies can be regarded to be the highly energy effective and offer the least data transfer delay in comparison with the remaining policies. These two optimal policies are considered to be the base; optimal one employing the 3G and Wi-Fi policy and another optimal one employing the Wi-Fi-only policy.

Employing Wi-Fi only policy with AWNIS, AWNIS-PSO and AWNIS-ABC: This policy utilizes Wi-Fi only for transferring the data. But, this policy carries out dynamic network interface selection using AWNIS and AWNIS-PSO. The network interface-selection interval varies in accordance with the environment the network is in.

Utilizing both 3G and Wi-Fi policy with AWNIS, AWNIS-PSO and AWNIS-ABC: This policy employs both 3G and Wi-Fi during data transfer and carries out dynamic network interface selection by means of AWNIS and AWNIS-PSO.

As observed in Fig. 3, the performance achieved of AWNIS, AWNIS-PSO and the newly introduced AWNIS-ABC algorithm differs based on the coefficient value. In case, the coefficient value equals to 0.1, then AWNIS-ABC algorithm takes up lesser energy of about 1.75 (J/Seconds). This indicates that the newly introduced AWNIS-ABC algorithm operates better in comparison with AWNIS-PSO and AWNIS algorithm.

As observed in Fig. 4, the performance of AWNIS, AWNIS-PSO and AWNIS-ABC algorithm differs based on the coefficient value α. If it was 0.1, AWNIS-PSO exhibited the greatest performance in terms of delay of 1.86 s only, while the other techniques like AWNIS-PSO and AWNIS generates delay of 1.93 and 2.12 s correspondingly.

Fig. 5 illustrates the average energy consumed in accordance with the anticipated amount of requested data transfer, where the number of APs is maintained at 15 and the static is kept at 30 s for the

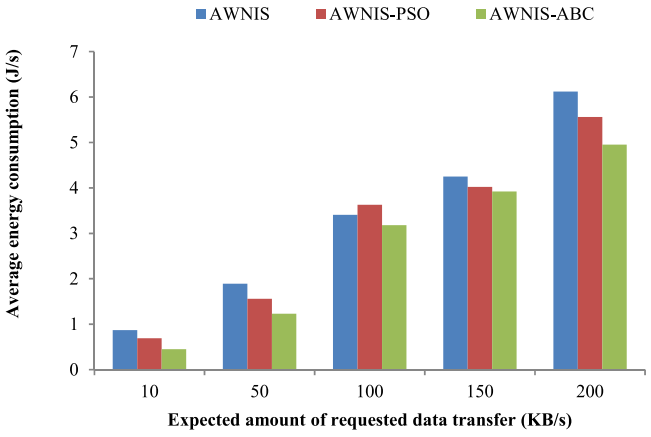


Fig. 5. Average energy consumption comparison as per the anticipated amount of requested data transfer where the number of APs is maintained 15 and for the policies with no AWNIS is 30 s.

policies with no AWNIS. As per Fig. 5, the average amount of energy consumption sees an increase as the increases. The energy consumption of 4G-only policy is considered the highest. As seen from Fig. 5, it can be concluded that the newly introduced AWNIS-ABC takes 4.95 J/S energy for 200 kB/seconds that is 0.61 J/S and 1.17 J/S lesser in comparison with AWNIS-PSO and AWNIS methods correspondingly. It is concluded that the newly introduced AWNIS-ABC exhibits superior performance compared to the policies AWNIS and AWNIS-PSO.

The ECG workload is an actual workload, which transmits 270 kB of sensed data to a surveillance server during each reporting time. This workload is identical to the periodical reporting of the status of

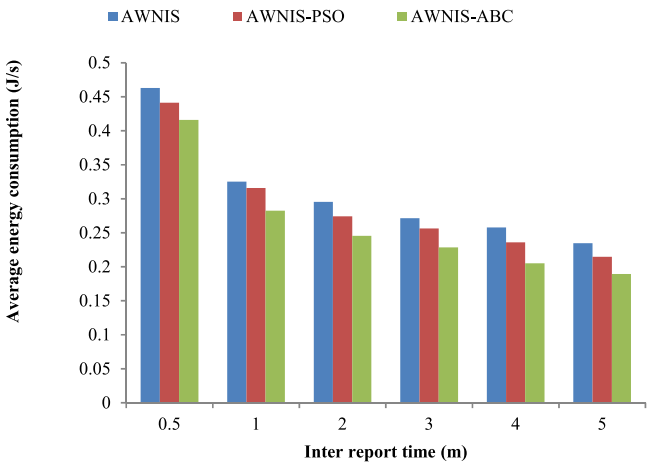


Fig. 6. Average energy consumption between methods as per the inter-report time with the number of APs is 15.

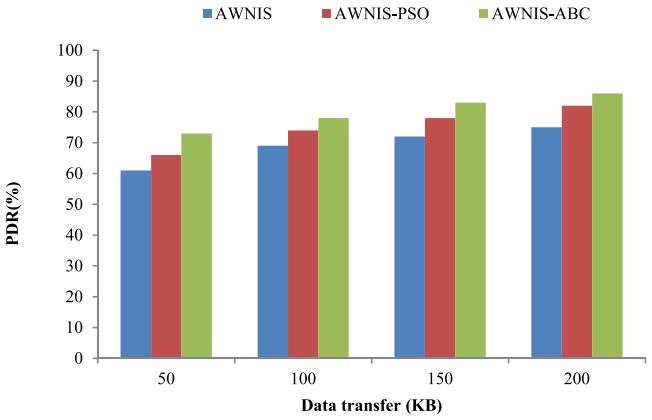


Fig. 7. PDR comparison vs. data transfer (methods).

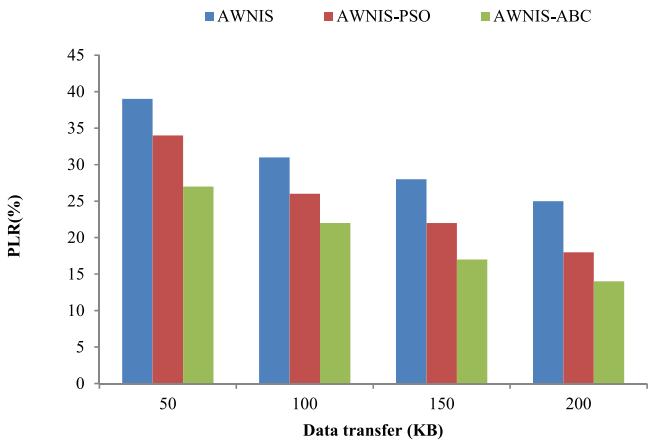


Fig. 8. PLR comparison vs. Data transfer (methods).

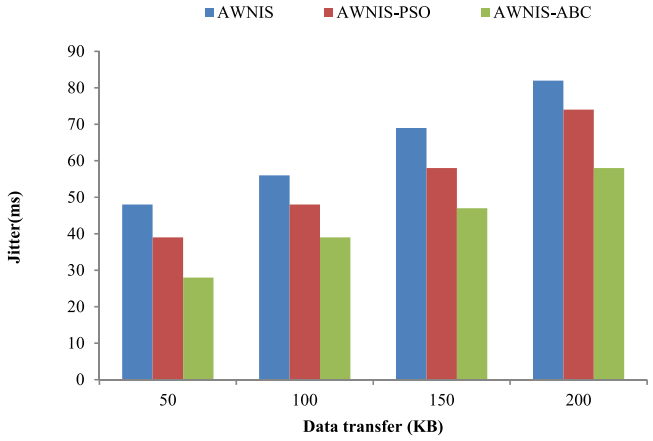


Fig. 9. Jitter comparison vs. Data transfer (methods).

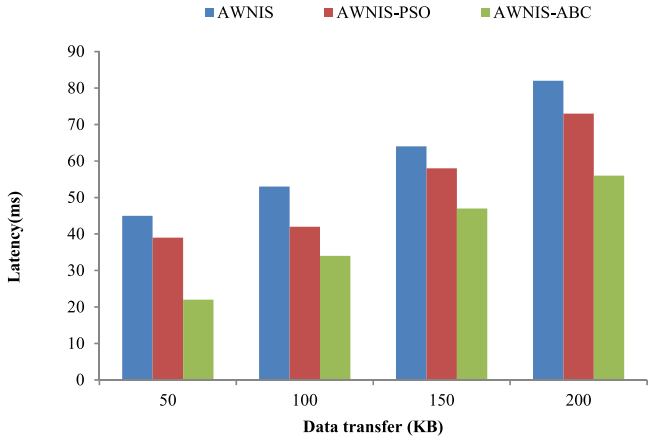


Fig. 10. Latency comparison vs. Data transfer (methods).

a machine in an industrial environment. Fig. 6 illustrates the average energy consumption and delay incurred between AWNIS, AWNIS-PSO and the newly introduced AWNIS-ABC. As illustrated in Fig. 6, the energy consumption of AWNIS-ABC is less compared to AWNIS and AWNIS-PSO if the inter-report time has a considerably greater value. As observed from Fig. 5 it can be determined that the energy consumed of the proposed AWNIS-ABC is about 0.1894 J/S energy for five minutes that is 0.0251 J/S and 0.0451 J/S lesser in comparison with AWNIS-PSO and AWNIS techniques correspondingly. It can be concluded that the newly introduced AWNIS-ABC exhibit superior performance compared to the policies AWNIS and AWNIS-PSO. In brief, when the inter-report time is considerably small, AWNIS-ABC technique offers more efficiency compared to AWNIS.

Fig. 7 illustrates the PDR results of the methods such as AWNIS, AWNIS-PSO and the proposed AWNIS-ABC algorithm. As illustrated in Fig. 7, the PDR of AWNIS-ABC is higher when compared to AWNIS and AWNIS-PSO. The proposed AWNIS-ABC algorithm gives higher PDR results of 86%, whereas other methods such as AWNIS and AWNIS-PSO gives lesser PDR results of 75% and 82% respectively for 200 kB.

Fig. 8 shows the PLR results of AWNIS, AWNIS-PSO and the proposed AWNIS-ABC algorithm. As illustrated in Fig. 8, the PLR of proposed AWNIS-ABC algorithm is higher when compared to AWNIS and AWNIS-PSO algorithms. The proposed AWNIS-ABC algorithm gives lesser PLR results of 14%, whereas other methods such as AWNIS and AWNIS-PSO gives higher PLR results of 25% and 18% respectively for 200 kB.

Fig. 9 shows the comparison result of AWNIS, AWNIS-PSO and the proposed AWNIS-ABC algorithm with respect to jitter. As illustrated in Fig. 9, the jitter of proposed AWNIS-ABC algorithm is 58 ms for 200 kB, whereas other methods such as AWNIS and AWNIS-PSO takes more time of 76 ms and 66 ms respectively for 200 kB.

Fig. 10 shows the latency comparison result of AWNIS, AWNIS-PSO and the proposed AWNIS-ABC algorithm. As illustrated in Fig. 10, the latency of proposed AWNIS-ABC algorithm is 56 ms for 200 kB, whereas other methods such as AWNIS and AWNIS-PSO takes more latency time of 82 ms and 73 ms respectively for 200 kB.

5. Conclusion and future work

Mobile devices like Personal Digital Assistants (PDAs) and smart phones are extensively employed not just everyday but also in different industrial areas. Many of these mobile devices are equipped with several wireless network interfaces, like Bluetooth, 3G, 4G, and Wi-Fi. There is a significant amount of energy consumption during the transmission of data via wireless communication. Therefore, the current industrial mobile terminals are fitted with several wireless network interfaces, and the operation of these terminals is done by less battery power. These mobile terminals are employed for monitoring and controlling the machines in industrial situations. Hence, the effectiveness of power usage is an extremely critical challenge. An energy-efficient adaptive wireless network interface-selection approach, which is termed as Adaptive Wireless Network Interface Selection (AWNIS) with Artificial Bee Colony (ABC), is proposed. Hidden Semi Markov Model (HSMM) technique is introduced for reducing the energy consumed at every sensor node and extending the network lifespan while transferring data packets from a source to a destination node, when meeting the packet threshold limits. Quality of Service (QoS) satisfaction of wireless network interfaces in mobile devices has been also carryout in this work. The following metrics such as Jitter, Packet Delivery Ratio (PDR), Packet Loss Rate (PER), and Latency have been used for WNIS. Depending on the results of the analysis, the newly introduced AWNIS-ABC approach is suitable for industrial wireless networks as it dynamically adapts the network-selection interval to minimize the energy usage in accordance with the network scenarios. The results of simulation indicate that the newly introduced approach efficiently minimizes the amount of energy consumption when ensuring a particular degree of data transfer delay. In the future work, ABC is integrated with other optimization techniques like ant colony, Simulated Annealing (SA), Bat Algorithm (BA) to resolve the important challenges of ABC thereby maximizing the results pertaining to energy efficiency.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

S. ArunKumar: Conceptualization, Methodology, Software, Data curation, Writing - original draft. **B. Vinoth Kumar:** Visualization, Investigation, Supervision. **M. Pandi:** Supervision, Writing - review & editing.

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